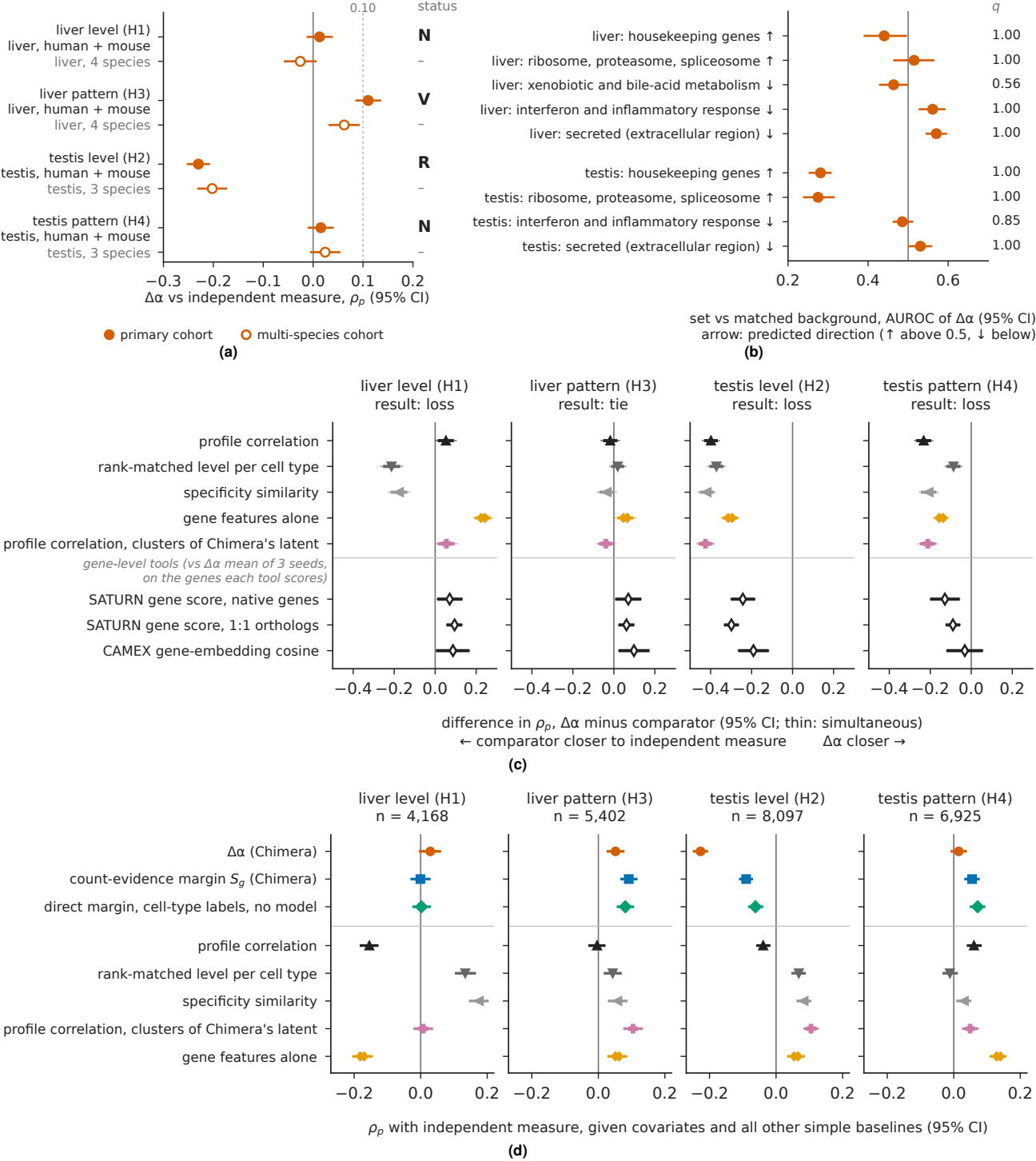

0.10
status
liver level (H1)
liver, human + mouse
liver, 4 species
liver pattern (H3)
liver, human + mouse
liver, 4 species
testis level (H2)
testis, human + mouse
testis, 3 species
testis pattern (H4)
testis, human + mouse
testis, 3 species
N
–
V
–
R
–
N
–
−0.3 −0.2 −0.1 0.0 0.1 0.2
Δα vs independent measure, ρ_p (95% CI)
primary cohort
multi-species cohort
(a)
q
liver: housekeeping genes ↑ 1.00
liver: ribosome, proteasome, spliceosome ↑ 1.00
liver: xenobiotic and bile-acid metabolism ↓ 0.56
liver: interferon and inflammatory response ↓ 1.00
liver: secreted (extracellular region) ↓ 1.00
testis: housekeeping genes ↑ 1.00
testis: ribosome, proteasome, spliceosome ↑ 1.00
testis: interferon and inflammatory response ↓ 0.85
testis: secreted (extracellular region) ↓ 1.00
0.2 0.4 0.6
set vs matched background, AUROC of Δα (95% CI)
arrow: predicted direction (↑ above 0.5, ↓ below)
(b)
liver level (H1)
result: loss
liver pattern (H3)
result: tie
testis level (H2)
result: loss
testis pattern (H4)
result: loss
profile correlation
rank-matched level per cell type
specificity similarity
gene features alone
profile correlation, clusters of Chimera's latent
gene-level tools (vs Δα mean of 3 seeds, on the genes each tool scores)
SATURN gene score, native genes
SATURN gene score, 1:1 orthologs
CAMEX gene-embedding cosine
−0.4 −0.2 0.0 0.2
difference in ρ_p, Δα minus comparator (95% CI; thin: simultaneous)
← comparator closer to independent measure Δα closer →
(c)
liver level (H1)
n = 4,168
liver pattern (H3)
n = 5,402
testis level (H2)
n = 8,097
testis pattern (H4)
n = 6,925
Δα (Chimera)
count-evidence margin S_g (Chimera)
direct margin, cell-type labels, no model
profile correlation
rank-matched level per cell type
specificity similarity
profile correlation, clusters of Chimera's latent
gene features alone
−0.2 0.0 0.2
ρ_p with independent measure, given covariates and all other simple baselines (95% CI)
(d)